(IEEE 830 / ISO 29148 Standard)

VQA-SRS-2026-V1 APPROVED BASELINE LEVEL 4 (CLOSED LOOP)

Autonomous Visual QA & Self-Healing Frontend Agent

End-to-End Patch & Verify
99.2%

Closed-Loop Self-Heal Cap
$0.03 / PR

SAM 3.1 Pixel Math Precision

Unit Compute & API Cost
Max 3x

< 15.0 sec

1. INTRODUCTION & SYSTEM SCOPE

1.1 Purpose: This Software Requirements Specification (SRS) defines the functional, mathematical, non-functional, and architectural constraints for VisorQA. Traditional DOM testing tools (Playwright, Cypress) verify node presence in the HTML tree but fail to detect visual clipping, z-index stacking occlusion, responsive grid breakages, and viewport bleed—costing engineering teams 25–30% of weekly sprint capacity.

1.2 Scope: VisorQA is an autonomous GitHub Action and FastAPI-orchestrated system that inspects Pull Requests across Desktop (`1440px`) and Mobile (`375px`) viewports inside isolated Modal sandboxes, detects layout defects via deterministic pixel math, localizes source files, synthesizes CSS/React patches, and commits verified fixes while streaming telemetry to a Live Mission Control UI.

2. SYSTEM ARCHITECTURE & ORCHESTRATION MODEL

Core Architectural Mandate: Pure Python Pydantic FSM Orchestrator (ADR-001)

The backend orchestrator (`orchestrator.py`) MUST be implemented as a Deterministic Finite State Machine (FSM) using Pure Python, `asyncio` , and `Pydantic` inside FastAPI. High-overhead multi-agent chat frameworks (LangChain, CrewAI, AutoGen) and full-repo Vector DB RAG pipelines are strictly prohibited. All state transitions mutate a single unified `AgentState` Pydantic model (`pr_url`, `attempt: 0..3`, `anomalies`, `target_files`, `repair_spec`, `current_patch`, `best_score`, `best_patch`, `is_resolved`) across five independent async nodes, broadcasting real-time events via `await ws.send_json(...)` .



3. FUNCTIONAL REQUIREMENTS (FR)

3.1 Event Trigger, Sandbox Provisioning & Vision Sensor (Sense Node)

Req ID	Module / Node	Deterministic System Requirement Specification
FR-01	Webhook & Modal Sandbox	The system SHALL autonomously ingest GitHub `pull_request` and `push` webhooks (< 1.2s latency), check out the target branch inside an isolated Modal container, and render the live application in headless Playwright across Desktop (`1440px`) and Mobile (`375px`) viewports.
FR-02	DOM-Guided SAM 3.1 Prompts	Playwright SHALL extract interactive DOM elements, assign each a unique `visor-el-id` , and pass its center point (c_x, c_y) and theoretical bounding box B_{DOM} as spatial prompts to SAM 3.1 to generate true visible pixel masks M_{SAM} with 1:1 element binding (prohibiting unprompted 64×64 grid

3.2 Triage, Architectural Diagnosis, Self-Healing Loop & Telemetry

Req ID	Module / Node	Deterministic System Requirement Specification
FR-04	LCA Extraction & Muse Spark Triage	For colliding or clipped elements, Playwright SHALL extract their Lowest Common Ancestor (LCA Parent Container), computed CSS (position , z index , flex , overflow), and repo file tree. This payload SHALL be routed once per run to Muse Spark via Baseten (2M TPM) to filter intentional overlaps (e.g., avatar notification badges) and isolate the 1–2 parent layout files (target_files), followed by broadcasting TRIAGE_COMPLETE .
FR-05	Muse Wonder Architect Diagnosis	Inside the retry loop (attempt < 3), Muse Wonder SHALL ingest the red SAM 3.1 collision mask overlay, math telemetry, and targeted source files, returning a structured Pydantic JSON repair_spec via the standard OpenAI SDK (response_format) without writing raw code.
FR-06	Muse Code Patch & Modal HMR	Muse Code SHALL synthesize a minimal Unified Diff Patch (current_patch) from repair_spec and apply it live inside the running Modal container via Hot-Module Replacement (HMR, 0.3s reload) without restarting the application server.
FR-07	Dual-Viewport Verification & 3-Strike Loop	The Verify Node SHALL re-execute SAM 3.1 across both Desktop (1440px) and Mobile (375px). If score > best_score , the state SHALL update best_score and best_patch . Resolution ($V_r \geq 0.98$ and IoU == 0 across both viewports with len(new_anomalies) == 0) immediately sets is_resolved = True and breaks the loop. Otherwise, new_anomalies feeds back into the next iteration and broadcasts ATTEMPT_{n}_FEEDBACK .
FR-08	Git Commit & Review Fallback	If is_resolved == True within 3 attempts, the system SHALL push a verified commit with before/after visual diffs to the PR branch. If 3 attempts fail to reach full convergence, the loop SHALL halt and post best_patch as a non-destructive suggested GitHub PR review comment.

4. NON-FUNCTIONAL REQUIREMENTS (NFR) & SYSTEM BENCHMARKS

NFR ID	Quality Attribute	Target Benchmark	Architectural Enforcement Mechanism
NFR-01	End-to-End Cycle Latency	< 15.0 seconds	Webhook trigger (<1.2s) + DOM-prompted SAM 3.1 (0.45s) + Muse Spark triage (0.6s) + Modal HMR reload (0.3s / 1.4s verify cycle).
NFR-02	Vision Accuracy & Speedup	99.2% / 10.6x speedup	Restricting SAM 3.1 prompts to 25–40 interactive DOM centers instead of 4,096 grid points cuts vision latency from 4.8s to 0.45s with zero VLM spatial hallucination.
NFR-03	Source Localization & Tokens	96.5% accuracy / -88.5% tokens	LCA DOM hierarchy + Muse Spark prunes context from ~28,000 tokens (10–15 files) to ~3,200 tokens (2 targeted files) and eliminates 40s Vector DB cold starts.
NFR-04	Cross-Page Side-Effect Rate	< 2.0% regression rate	Dual-viewport verification + shared atomic UI component protection boosts first-pass fix rate from 54% to 91%.
NFR-05	Unit Economics & Cost Cap	$0.03 / Pull Request	Hard max_attempts = 3 cap and context pruning ensure commercial viability under the $199–$499/month B2B SaaS tier.

5. TECHNICAL CONSTRAINTS & BUSINESS RULES

5.1 Technical Constraints (Strict Architectural Guardrails)

1. **TC-1 (Zero Agent Frameworks):** The backend **MUST NOT** use LangChain, CrewAI, or AutoGen. Orchestration must be written in **Pure Python + Asyncio + Pydantic** inside `orchestrator.py` to prevent conversational token burn, latency overhead, and NumPy/WebSocket serialization bottlenecks.
2. **TC-2 (Zero Vector DB / No Full-Repo RAG):** The system **MUST NOT** index repositories into Vector Databases. File localization relies strictly on DOM Lowest Common Ancestor (LCA) extraction + Muse Spark, saving ~40s of cold-start indexing and 70% container memory.
3. **TC-3 (Zero VLM Spatial Guessing & No SAM Grid Mode):** Vision LLMs must never guess pixel coordinates, and SAM 3.1 must never run in unprompted 64×64 "Everything Mode". All bug detection is 100% deterministic via DOM-guided SAM 3.1 masks and NumPy math.
4. **TC-4 (Architect-Coder Separation): Muse Wonder** is restricted to outputting JSON `repair_spec` diagnostics; **Muse Code** is restricted to generating Unified Diff patches from that spec. Single-shot VLM code rewriting is prohibited.

5.2 Business Rules (Autonomous Governance & Safety)

1. **BR-1 (The 3-Strike Closed-Loop Cap):** Self-healing execution is strictly capped at `max_attempts = 3`. If $V_r \geq 0.98$ and IoU = 0 are not achieved by Strike 3, the loop halts immediately to prevent infinite token burn, and the highest-scoring patch (`best_patch`) is posted as a suggested GitHub PR review comment rather than a direct commit.
2. **BR-2 (Shared Design-System Protection):** Atomic UI library components (`src/components/ui/*`, e.g., `Button.tsx`) are read-only by default during triage. Mutations must target the parent layout container (e.g., `CheckoutPage.tsx`) or colliding sibling (`PromoBanner.tsx`) to keep cross-page side-effect rates < 2%.
3. **BR-3 (Cross-Viewport Zero-Regression Rule):** A mobile CSS fix (375px) **MUST NEVER** be committed if it degrades desktop (1440px) visibility below $V_r \geq 0.98$ or introduces sibling collisions (IoU > 0). Both viewports must pass simultaneously.
4. **BR-4 (Anti-Hack CSS Policy):** Generated patches must resolve layout structure (e.g., flex-wrap, container overflow, contextual stacking) and are prohibited from injecting brittle global overrides (`!important` or arbitrary `z-index: 9999`).
5. **BR-5 (Unit Economics Guardrail):** Total compute and API inference cost must remain bounded at an average of **$0.03 per PR** to maintain commercial viability for the $199–$499/month B2B SaaS tier.